Aryan Naik

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EDUCATION

MEng Robotics and Artificial Intelligence | University College London (UCL) | 2025 - 2029

Second year. Core modules span embedded engineering, control theory, machine learning, and artificial intelligence. Relevant coursework includes sensor systems, mechatronics, and autonomous systems design.

PROJECTS

Aerospace Avionics Firmware — Student Rocketry Telemetry Payload | ALRS 007 Capsule

- Delivered a flight-ready avionics firmware system across a dual-MCU embedded architecture, enabling autonomous parachute deployment and descent control under safety-critical flight conditions via a finite state machine with interrupt-driven real-time execution
- Integrated IMU, altimeter, and accelerometer sensors over I2C serial bus, producing continuous inertial measurement, attitude estimation, and altitude logging across the full flight envelope
- Maintained ground visibility of vehicle state by implementing wireless telemetry transmission and onboard flash data logging, capturing sensor data at consistent intervals throughout ascent and descent phases
- Applied hardware-software co-design principles including power budgeting and signal conditioning to meet embedded resource constraints on a mass- and volume-limited flight computer

Keywords: avionics, rocketry, guidance navigation and control, autonomous recovery systems, safety-critical embedded software, real-time firmware, UART, SPI, I2C, sensor fusion, defence electronics

Full-System Robotics Hardware Integration | Robotics Challenge

- Delivered a complete functional robotic platform under competition constraints by integrating microcontrollers, motor drivers, actuators, and power electronics on breadboard from schematic to physical assembly
- Verified system reliability through systematic fault-finding and hardware debugging, resolving signal integrity faults across sensor, actuator, and power subsystems prior to acceptance testing
- Achieved closed-loop sensor-actuator feedback control by developing microcontroller firmware for real-time servo and H-bridge motor control driven by ultrasonic and IR distance sensor inputs
- Produced all structural enclosures and mounting geometry in Autodesk Fusion 360, applying tolerancing and clearance constraints to ensure correct component fit across the assembled platform

Keywords: mechatronics, electronics engineering, PCB layout, voltage regulation, signal conditioning, rapid prototyping, systems integration, CAD, 3D printing

SKILLS

Embedded and Firmware: C++, real-time firmware, finite state machine design, interrupt-driven execution, UART, SPI, I2C, microcontroller programming, sensor fusion, power budgeting, signal conditioning

Systems and Hardware: hardware-software co-design, breadboard electronics, circuit wiring and debugging, motor control, servo control, voltage regulation, component selection, systems integration and acceptance testing

Design and CAD: Autodesk Fusion 360, mechanical design, structural enclosures, tolerancing, 3D printing, rapid prototyping

Aerospace and Defence: avionics firmware (student rocketry project), safety-critical embedded software, guidance navigation and control, autonomous recovery systems, flight computer design (student rocketry project), defence electronics awareness

Engineering Disciplines: aerospace engineering, mechatronics, electronics engineering, embedded systems engineering

ADDITIONAL

Available for summer 2027 internship. Eligible to work in the UK without visa sponsorship.

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Hiring Team Boeing Prestwick / Farnborough / Welwyn Garden City

Dear Hiring Team,

Boeing's position at the intersection of civil aviation, defence systems, and space technology makes it one of the most technically demanding environments an engineering student could enter. What draws me specifically is Boeing's commitment to safety-critical systems engineering at scale, from flight computer architectures on commercial aircraft to guidance and navigation on defence platforms. As a second-year MEng Robotics and Artificial Intelligence student at University College London, I am eager to apply my embedded systems and aerospace engineering experience within that context during Summer 2027.

My most directly relevant work is the ALRS 007 Capsule, an aerospace avionics payload I developed for a student rocketry programme. I designed and implemented C++ firmware on a dual-MCU embedded architecture, integrating IMU, altimeter, and accelerometer sensors over I2C to perform real-time attitude estimation and flight data logging. The system used a finite state machine to autonomously manage parachute deployment under safety-critical flight conditions, with redundant onboard flash logging and wireless telemetry transmission. The result was a fully functional flight computer that successfully executed autonomous recovery across live test conditions, demonstrating reliable performance in a constrained, high-stakes embedded environment.

At Boeing, I would contribute practical experience in real-time firmware development, sensor integration, and safety-critical embedded software design, skills directly applicable to avionics and defence electronics work. I am keen to develop my understanding of how aerospace engineering principles are applied at an industrial scale, particularly in areas such as systems integration, verification and validation, and the rigorous certification processes that govern flight-critical hardware. I am confident this placement would be a formative step in my engineering career.

Yours sincerely, Aryan Naik